

COMPLEX NUMBERS

Real Numbers

$\sqrt{-ve} \rightarrow$ Real Numbers $\rightarrow \sqrt{+ve}$

Rational

Irrational

Natural

Whole

Integers.

$$b^2 - 4ac > 0 \rightarrow (+ve)$$

$$b^2 - 4ac = 0 \rightarrow 0$$

$$* ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

* $D = b^2 - 4ac > 0 \rightarrow$ Two, different solⁿ

* $D = b^2 - 4ac = 0 \rightarrow$ Two, Same and Real solⁿ

* $D = b^2 - 4ac < 0 \rightarrow$ No Real solⁿ (-ve).

Decimal Expansion

Decimal Expansion

Terminating

Non-terminating Repeating

Non-terminating Non-Repeating

Complex Number

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$x = \sqrt{-1} \rightarrow \text{Imaginary Root}$$

$$* \sqrt{-1} = i \rightarrow i^2 = -1$$

$$* i^2 = (\sqrt{-1})^2 = -1$$

Complex Number

$$* z = a + ib \text{ where } a \text{ \& } b \text{ are real no.}$$

$\downarrow$
Complex number

$$-a = +(-a)$$

$$\begin{array}{c} \oplus \\ 2 \end{array} - \begin{array}{c} \ominus \\ 5 \end{array} \neq -\begin{array}{c} \oplus \\ 2 \end{array} + \begin{array}{c} \ominus \\ 5 \end{array}$$

$$\text{Eg} \rightarrow 2 + i3, \frac{1}{2}, i\sqrt{2}, \left(-\frac{3}{2}\right) + i5$$

$$\text{Re } z = 2$$

$$\text{Im } z = 3$$

$$\begin{array}{c} \ominus \\ i \end{array} \begin{array}{c} \oplus \\ 2 \end{array} = \begin{array}{c} \oplus \\ 1 \end{array} + \begin{array}{c} \ominus \\ i \end{array} \begin{array}{c} \oplus \\ (-2) \end{array}$$

$\downarrow$ $\downarrow$
Re Im

Complex Number

$$z = a + i(b) \rightarrow \text{Imaginary part of complex no. } z$$

$\downarrow$

Real part of complex no. z
($\text{Re } z$) = a

Complex Number

$$z_1 = a + ib$$

$$\text{if } a = c$$

$$\& \quad b = d$$

Then

$$z_1 = z_2$$

$$z_1 = 2 + i(5) \quad z_2 = \frac{4}{2} + i\left(\frac{10}{2}\right)$$

$$z_1 = z_2$$

$$z_2 = c + id$$

$$z_1 = 2 + i(5) \quad z_2 = 2 + i(5) = 2 + i(-5)$$

$$z_1 \neq z_2$$

$$2 = 2$$

$$5 \neq -5$$

Complex Number

$\sqrt{-1} \rightarrow$ Imaginary number

$$* \quad i = \sqrt{-1}$$

$$* \quad i^2 = -1$$

$$* \quad z = a + i(b) \rightarrow \text{Im } z$$

$\downarrow$

Re z

$$* \quad z_1 = a_1 + ib_1, \quad z_2 = a_2 + ib_2$$

$$\text{if } a_1 = a_2$$

$$b_1 = b_2$$

Then,

$$z_1 = z_2$$

Important points

- (i) Sum of a rational and irrational is irrational.
- (ii) Difference of a rational and an irrational is irrational
- (iii) product of a rational and an irrational is irrational
 $\rightarrow$ (Except 0)
- (iv) Quotient of a rational and an irrational is irrational
 $\rightarrow$ (Except 0)

Important points

- (i) Sum of two irrationals need not be an irrational.
- (ii) Difference of two irrationals need not be an irrational.
- (iii) Product of two irrationals need not be an irrational.
- (iv) Quotient of two irrationals need not be an irrational.

Complex Number

* Algebra of Complex numbers: -

* Addition of two complex numbers; -

$$z_1 = a + ib, \quad z_2 = c + id$$

$$z_1 + z_2 = a + ib + c + id$$

$$z_1 + z_2 = a + c + i(b + d)$$

$$\boxed{z_1 + z_2 = (a + c) + i(b + d)}$$

$$\underline{\underline{Q}} \quad z_1 = 2 + i3, \quad z_2 = 5 + i7$$

$$\Rightarrow z_1 + z_2 = (2 + 5) + i(3 + 7) \\ = 7 + i(10)$$

$$\underline{\underline{Q}} \quad z_1 = \left(-\frac{1}{2}\right) + i2, \quad z_2 = 2 + i3$$

$$z_1 + z_2 = \left(-\frac{1}{2} + 2\right) + i(2 + 3) \\ = \frac{3}{2} + i5$$

$$\underline{\underline{Q}} \quad z_1 = 2 + i5, \quad z_2 = 3 - i2$$

$$\rightarrow z_1 = 2 + i5 \\ z_2 = 3 - i2 = 3 + i(-2)$$

$$z_1 + z_2 = (2 + 3) + i(5 + (-2)) \\ = 5 + i3$$

$$\underline{\underline{Q}} \quad z_1 = -2 + i5, \quad z_2 = 3 - i2$$

$$\rightarrow z_1 = (-2) + i5 \\ z_2 = 3 + i(-2)$$

$$z_1 + z_2 = (-2 + 3) + i(5 + (-2)) \\ = 1 + i(-2) \\ = 1 - i2$$

Complex number

* The closure law $\rightarrow z_1 + z_2 = \textcircled{z} \rightarrow$ complex numbers
 $\downarrow \quad \downarrow$
complex numbers

* The commutative law $\rightarrow z_1 + z_2 = z_2 + z_1$

* The Associative law $\rightarrow (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

Complex Number

* Additive Identity $\rightarrow 0 + i0 = 0$
 $\downarrow$
Zero complex number

$$\boxed{z + 0 = z}$$

eg $\rightarrow z = a + ib$
 $0 = 0 + i0$

$$z + 0 = (a + i0) + i(b + i0)$$
$$z + 0 = a + ib$$

$$\boxed{z + 0 = z}$$

* Additive Inverse $\rightarrow$

$$z = a + ib$$

$$\boxed{(-a) + i(-b) = (-z)}$$

$$= (a + (-a)) + i(b + (-b))$$
$$= 0 + i(0)$$

$$\boxed{z + (-z) = 0}$$

Complex Number

* Algebra of Complex number $\rightarrow$

* Difference of two Complex number $\rightarrow$

$$z_1 = a + ib$$

$$z_2 = c + id, -z_2 = -c - id$$

$$z_2 = -c + i(-d)$$

$$z_1 - z_2 = z_1 + (-z_2)$$

$$= (a + ib) + (-c) + i(-d)$$

$$\boxed{z_1 - z_2 = (a - c) + i(b - d)}$$

$$\underline{\text{Ex}} \quad z_1 = 2 + i3, \quad z_2 = 5 + i7$$

$$\begin{aligned} \Rightarrow z_1 - z_2 &= (2-5) + i(3-7) \\ &= -3 + i(-4) \\ &= -3 - i4 \end{aligned}$$

$$\underline{\text{Q}} \quad z_1 = \left(-\frac{1}{2}\right) + i2, \quad z_2 = 2 + i3$$

$$\Rightarrow -z_2 = -2 - i3 = -2 + i(-3)$$

$$\begin{aligned} z_1 - z_2 &= z_1 + (-z_2) = \left[\left(-\frac{1}{2}\right) + (-2) \right] + i(2 + (-3)) \\ &= -\frac{5}{2} + i(-1) \\ &= -\frac{5}{2} - i \end{aligned}$$

$$* \quad i = \sqrt{-1}$$

$$* \quad i^2 = -1$$

$$* \quad z = a + ib \rightarrow \begin{array}{l} \uparrow \\ \text{Real part} \\ (\text{Re } z) \end{array} \quad \begin{array}{l} \text{Imaginary} \\ \text{part (Im } z) \end{array}$$

$$* \quad z_1 = a + ib, \quad z_2 = c + id$$

$$\text{if } a=c, \quad b=d$$

$$z_1 = z_2$$

Algebra of Complex Number

• Addition of two Complex Number:-

$$z_1 = a + ib, \quad z_2 = c + id$$

$$\boxed{z_1 + z_2 = (a+c) + i(b+d)}$$

$$* \quad z_1 + z_2 = z_2 + z_1$$

$$* \quad (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

$$* \quad 0 + i0 = 0, \quad (z + 0) = z$$

$$* \quad a + ib + (-a) + i(-b) = 0 + i0 = 0$$

• Difference of two Complex Number -

$$z_1 = a + ib, \quad z_2 = c + id \Rightarrow -z_2 = -c + i(-d)$$

$$\begin{aligned} z_1 - z_2 &= (a + ib) - (c + id) \\ &= a + ib - c - id \end{aligned}$$

$$\boxed{z_1 - z_2 = (a-c) + i(b-d)}$$

$$* \quad z_1 - z_2 = z_1 + (-z_2)$$

$$= (a + (-c)) + i(b + (-d))$$

$$= (a-c) + i(b-d)$$

Algebra of Complex Numbers

• Multiplication of two Complex numbers -

$$\begin{aligned}z_1 &= a+ib \quad z_2 = c+id \\z_1 \times z_2 &= (a+ib) \times (c+id) \\&= ac + iad + ibc + i^2 bd \\&= ac + i(-1)bd + i(ad+bc)\end{aligned}$$

$$\boxed{z_1 \times z_2 = (ac - bd) + i(ad + bc)}$$

Q $z_1 = 2+i7 \quad z_2 = 3-i4 = z_1 \times z_2$

$$\begin{aligned}z_1 \times z_2 &= (2+i7) \times (3+i(-4)) \\&= 6 - 8i + i21 - i^2 28 \\&= 6 + i(-8+21) - (-1)28 \\&= 6 + i13 + 28\end{aligned}$$

$$\boxed{z_1 \times z_2 = 34 + i13}$$

* $z_1 \times z_2 = z \rightarrow$ complex number

$\downarrow \downarrow$

Complex number

* $z_1 \times z_2 = z_2 \times z_1$

* $(z_1 \times z_2) \times z_3 = z_1 \times (z_2 \times z_3)$

* Multiplicative identity $\rightarrow 1+i0 = 1, z \times 1 = z$

* Multiplicative Inverse $\rightarrow z = a+ib$

$$\frac{1}{z} = \left(\frac{a}{a^2+b^2} \right) + i \left(\frac{-b}{a^2+b^2} \right) = z^{-1}$$

$$\boxed{z \times \frac{1}{z} = 1}$$

* $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Algebra of Complex Numbers

• Division of two Complex Numbers

$$z_1 = a + ib, z_2 = c + id \Rightarrow \frac{1}{z_2} = \frac{c}{c^2 + d^2} + i \left(\frac{-d}{c^2 + d^2} \right)$$

$$\frac{z_1}{z_2} = z_1 \times \frac{1}{z_2} = (a + ib) \left(\frac{c}{c^2 + d^2} + i \left(\frac{-d}{c^2 + d^2} \right) \right)$$

Q $z_1 = 5 + i7, z_2 = 2 - i3, \frac{z_1}{z_2}, \frac{z_2}{z_1}$
 $\Rightarrow z_1 = 5 + i7, z_2 = 2 - i3 = 2 + i(-3)$

$$\boxed{\frac{z_1}{z_2} = z_1 \times \frac{1}{z_2}}$$

$$\frac{1}{z_2} = \left(\frac{2}{(2)^2 + (-3)^2} + i \left(\frac{-(-3)}{(2)^2 + (-3)^2} \right) \right)$$

$$\frac{1}{z_2} = \left(\frac{2}{13} + i \frac{3}{13} \right)$$

$$\frac{z_1}{z_2} = z_1 \times \frac{1}{z_2} = (5 + i7) \times \left(\frac{2}{13} + i \frac{3}{13} \right)$$

$$= \frac{10}{13} + i \frac{15}{13} + i \frac{14}{13} + i^2 \frac{21}{13}$$

$$= \frac{10}{13} + i \left(\frac{15}{13} + \frac{14}{13} \right) + (-1) \frac{21}{13}$$

$$= \frac{10}{13} - \frac{21}{13} + i \left(\frac{15 + 14}{13} \right)$$

$$= \frac{10 - 21}{13} + i \left(\frac{29}{13} \right)$$

$$\boxed{\frac{z_1}{z_2} = \frac{-11}{13} + i \left(\frac{29}{13} \right)}$$

$$\frac{z_2}{z_1} = z_2 \times \frac{1}{z_1}$$

$$z_1 = 5 + i7$$

$$\frac{1}{z_1} = \frac{5}{25 + 49} + i \frac{-7}{25 + 49}$$

$$\frac{1}{z_1} = \frac{5}{74} + i \left(\frac{-7}{74} \right)$$

$$\frac{z_2}{z_1} = (2 + i(-3)) \times \left(\frac{5}{74} + i \left(\frac{-7}{74} \right) \right)$$

$$= \frac{10}{74} + i \left(\frac{-2 \times 7}{74} \right) + i \left(\frac{-3 \times 5}{74} \right)$$

$$+ i^2 \left(-3 \times \left(\frac{-7}{74} \right) \right)$$

$$\frac{z_2}{z_1} = \frac{5}{37} + i \left(\frac{-7}{37} - \frac{15}{74} \right) + (-1) \left(\frac{21}{74} \right)$$

$$= \frac{5}{37} - \frac{21}{74} + i \left(\frac{-14 - 15}{74} \right)$$

$$= \frac{10 - 21}{74} + i \left(\frac{-29}{74} \right)$$

$$\boxed{\frac{z_2}{z_1} = \frac{-11}{74} - i \left(\frac{29}{74} \right)} \quad \text{Ans.}$$

Complex Number

* Power of $i \rightarrow$

$$* i = \sqrt{-1}$$

$$* i^2 = -1$$

$$* i^3 = i^2 \times i = (-1) \times i$$

$$\boxed{i^3 = -i}$$

$$* i^4 = i^2 \times i^2 = (-1) \times (-1)$$

$$\boxed{i^4 = 1}$$

$$* i^5 = i^3 \times i^2 = -i \times (-1)$$

$$\boxed{i^5 = i}$$

$$* i^{-1} = \frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1}$$

$$\boxed{i^{-1} = -i}$$

$$* i^{-4} = \frac{1}{i^4} = \frac{1}{1}$$

$$\boxed{i^{-4} = 1}$$

$$* i^{-2} = \frac{1}{i^2} = \frac{1}{-1}$$

$$\boxed{i^{-2} = -1}$$

$$* i^{-3} = \frac{1}{i^3} = \frac{1}{i^2 \times i} = \frac{1}{-1 \times i} = \frac{-1}{i} = -i^{-1} = -(-i)$$

$$\boxed{i^{-3} = i}$$

Algebra of Complex Number

• Addition of two Complex Number.

$$z_1 = a + ib, z_2 = c + id$$

$$z_1 + z_2 = (a + c) + i(b + d)$$

• Difference of two Complex Number.

$$z_1 = a + ib, z_2 = c + id \Rightarrow -z_2 = -c + id = -c + i(-d)$$

$$z_1 - z_2 = z_1 + (-z_2)$$

$$= (a - c) + i(b - d)$$

• Multiplication of two Complex Number $\rightarrow$

$$z_1 = a + ib, z_2 = c + id$$

$$z_1 z_2 = (a + ib)(c + id)$$

$$= ac + iad + ibc + (i)^2 bd$$

$$= ac + (-1)bd + i(ad + bc)$$

$$= (ac - bd) + i(ad + bc)$$

$$z = a + ib$$

$$\boxed{\frac{1}{z} = z^{-1} = \frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2}}$$

Algebra of Complex Number

• Division of two Complex Numbers -

$$z_1 = a + ib \quad z_2 = c + id$$

$$\frac{z_1}{z_2} = z_1 \times \frac{1}{z_2} = (a + ib) \times \left(\frac{c}{c^2 + d^2} + i \frac{-d}{c^2 + d^2} \right)$$

Complex Number

$$* i = \sqrt{-1}$$

$$* i^2 = -1$$

$$* i^3 = i^2 \cdot i = (-1) \cdot i$$

$$i^3 = -i$$

$$* i^4 = (i^2)^2 = (-1)^2$$

$$i^4 = 1$$

$$* i^5 = i^3 \cdot i^2 = (-i) \cdot (-1)$$

$$i^5 = i$$

$$* i^6 = (i^2)^3 = (-1)^3$$

$$i^6 = -1$$

Complex Number

* Square Root of negative Real number -

$$* \boxed{\sqrt{-1} = i}$$

$$i^2 = -1$$

$$(-i)^2 = i^2 = -1$$

$$\boxed{-i = \sqrt{-1}}$$

* Square Root of negative Real numbers

$$* \sqrt{-2} = \sqrt{2 \times (-1)} = \sqrt{2} \times \sqrt{-1}$$

$$\boxed{\sqrt{-2} = i\sqrt{2} = \sqrt{2}i}$$

$$* \sqrt{\frac{-3}{2}} = \sqrt{(-1) \times \frac{3}{2}} = \sqrt{-1} \times \sqrt{\frac{3}{2}}$$

$$\boxed{\sqrt{\frac{-3}{2}} = i\sqrt{\frac{3}{2}}}$$

Complex Numbers

$\sqrt{a} \rightarrow$ Real no.

* Square Root of negative Real numbers -

$$* \boxed{\sqrt{-a} = i\sqrt{a} = \sqrt{a}i}$$

Complex Number

$$* \boxed{\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}} \quad \left\{ \begin{array}{l} \text{where } a > 0 \text{ \& } b < 0 \\ \text{or} \\ a < 0 \text{ \& } b > 0 \end{array} \right.$$

$$\text{X} \rightarrow \boxed{a < 0, b < 0}$$

$$\boxed{i^2 = -1}$$

$$i^2 = i \times i = \sqrt{-1} \times \sqrt{-1} = \sqrt{(-1) \times (-1)} = \sqrt{1}$$

$$\boxed{i^2 = 1} \text{ X}$$

Complex Number

$$(z_1 \pm z_2)^2 = z_1^2 + z_2^2 \pm 2z_1 z_2$$

$$(z_1 \pm z_2)^3 = z_1^3 \pm z_2^3 \pm 3z_1 z_2 (z_1 \pm z_2)$$

$$z_1^2 - z_2^2 = (z_1 - z_2)(z_1 + z_2)$$

Complex Number

* Modulus and Conjugate of Complex Number -

$$z = a + ib$$

$$* \boxed{\text{Modulus of } z = |z| = |a + ib| = \sqrt{a^2 + b^2}}$$

$$* \boxed{\text{Conjugate of } z = \bar{z} = a - ib}$$

Complex Number

$$z = a + ib$$

$$\frac{1}{z} = z^{-1} = \frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2} = \frac{a + i(-b)}{a^2 + b^2}$$

$$\frac{1}{z} = \frac{a - ib}{a^2 + b^2} = \frac{\bar{z}}{|z|^2}$$

$$* \boxed{\frac{1}{z} = z^{-1} = \frac{\bar{z}}{|z|^2}}$$

Modulus of Complex Number

$$* \sqrt{-a} = i\sqrt{a}$$

$$* \sqrt{a \times b} = \sqrt{a} \times \sqrt{b} \quad \{ a < 0 \text{ \& } b > 0 \text{ \& } a > 0 \text{ \& } b < 0 \}$$

$$* z = a + ib$$

$$|z| = \sqrt{a^2 + b^2}$$

Conjugate of Complex Number

$$z = a + ib$$

$$\bar{z} = a - ib$$

$$* z^{-1} = \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

$$* |z_1 z_2| = |z_1| \times |z_2| \quad * \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$* \overline{z_1 z_2} = \bar{z}_1 \times \bar{z}_2 \quad * \overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2 \quad * \left(\frac{\bar{z}_1}{z_2} \right) = \frac{\bar{z}_1}{z_2}$$

Modulus of Complex Number

$$* \sqrt{-a} = i\sqrt{a}$$

$$* \sqrt{a \times b} = \sqrt{a} \times \sqrt{b} \quad \{ a < 0 \text{ \& } b > 0 \text{ \& } a > 0 \text{ \& } b < 0 \}$$

$$* z = a + ib$$

$$|z| = \sqrt{a^2 + b^2}$$

Conjugate of Complex Number

$$z = a + ib$$

$$\bar{z} = a - ib$$

$$* z^{-1} = \frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2}$$

$$* z^{-1} = \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

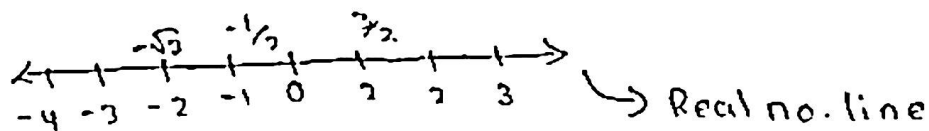
$$* |z_1 z_2| = |z_1| \times |z_2|$$

$$* \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$* \overline{z_1 z_2} = \bar{z}_1 \times \bar{z}_2$$

$$* \overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2 \quad * \left(\frac{\bar{z}_1}{z_2} \right) = \frac{\bar{z}_1}{z_2}$$

* Argand plane & polar Representation →

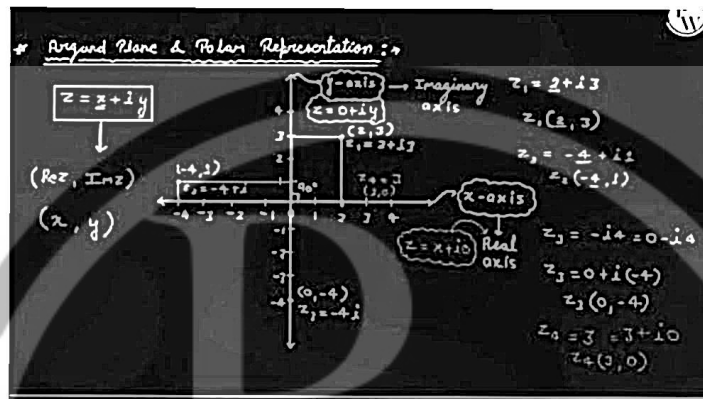


* Argand plane & polar Representation →

$$Z = x + iy$$

↓
(Re_Z, Im_Z)

(x, y)



$$Z_1 = 2 + i3$$

$$Z_1(2, 3)$$

$$Z_2 = -4 + i1$$

$$Z_2(-4, 1)$$

$$Z_3 = -i4 = 0 - i4$$

$$Z_3 = 0 + i(-4)$$

$$Z_3(0, -4)$$

$$Z_4 = 3 = 3 + i0$$

$$Z_4(3, 0)$$

* Argand plane & polar Representation →

$$Z = a + ib$$

$$Z(a, b)$$

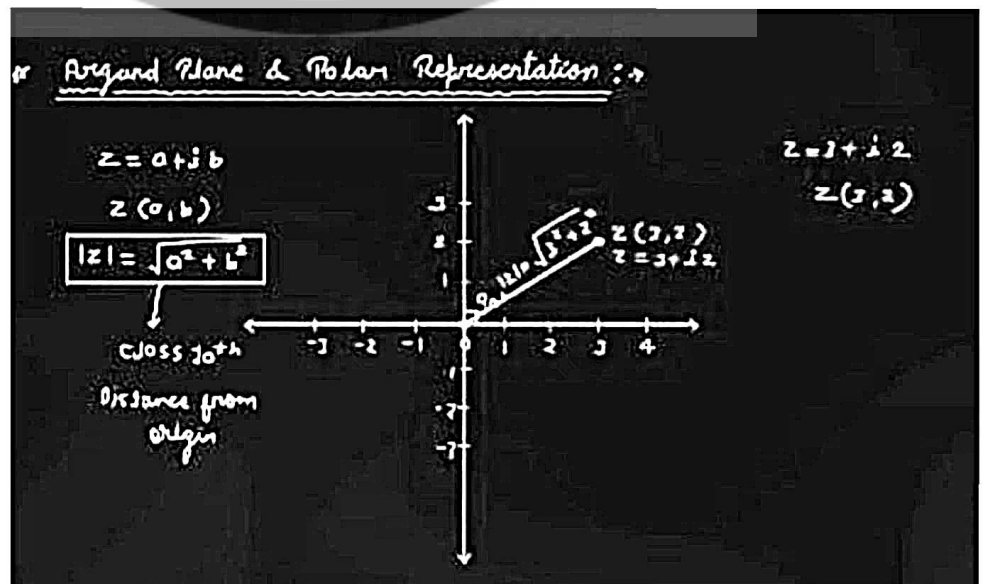
$$|Z| = \sqrt{a^2 + b^2}$$

↓
class 10th

Distance from
origin.

$$Z = 3 + i2$$

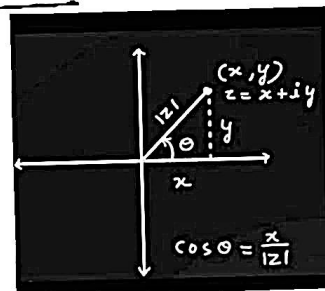
$$Z(3, 2)$$



* Argand plane & polar Representation $\Rightarrow$

$$* z = x + iy$$

$$z = |z| \cos \theta + i |z| \sin \theta$$



$$* \boxed{z = |z| (\cos \theta + i \sin \theta)} \rightarrow \text{polar Representation}$$

$$\boxed{\tan \theta = \frac{y}{x}}$$

$$* \boxed{\theta = \tan^{-1} \left(\frac{y}{x} \right)} \rightarrow \text{Argument of Complex no.}$$

$$\cos \theta = \frac{x}{|z|}$$

$$|z| \cos \theta = x$$

$$\sin \theta = \frac{y}{|z|}$$

$$|z| \sin \theta = y$$